

ASPIRE COLLEGE VEHARI CAMPUS

Notes • BS CS 3rd Semester • Lecture 7: Flow Control & Sliding Window Protocols

Course: Computer Networks (Credit Hours: 3)

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1. WHAT IS FLOW CONTROL? (SPEED MATCHING)

Flow Control is a Data Link Layer technique that coordinates the transmission speed of a fast sender with a slower receiver so the receiver's memory buffer never overflows.

Easy Everyday Example Pouring tea into a small teacup: If you pour too fast from the kettle, the tea spills over. Flow Control tells the sender to pause pouring until the cup is empty!

2. FLOW CONTROL APPROACHES

Flow control is divided into two primary categories:

1. Stop-and-Wait

Sender sends **1 frame** and halts.

- Waits for an ACK (Acknowledgment).
- Sends next frame only after ACK arrives.
- **Downside:** Very slow; bandwidth is mostly wasted sitting idle.

2. Go-Back-N (GBN)

Sender sends up to **N frames** continuously without waiting for individual ACKs.

- Uses a **Sliding Window**.
- If 1 frame is lost, receiver drops it and **all subsequent frames**.
- Sender re-sends all N frames starting from the lost one.

3. Selective Repeat

Sender transmits up to N frames. Receiver has extra buffer memory.

- If frame #3 is lost, receiver stores frame #4 & #5 safely.
- Sender only re-transmits **frame #3**.
- Much more efficient; saves bandwidth.

3. SUMMARY COMPARISON: GBN VS. SELECTIVE REPEAT

Feature	Go-Back-N (GBN)	Selective Repeat (SR)
Receiver Buffer	Size = 1 (discards out-of-order frames)	Size = N (can hold out-of-order frames)
Retransmission	Re-sends the lost frame + ALL following frames	Re-sends ONLY the specific lost or damaged frame
Complexity	Simple logic, less memory needed	More complex logic, requires memory buffers
Bandwidth Efficiency	Lower (wastes capacity on duplicate re-sends)	High (no redundant transmissions)

4. PIGGYBACKING TECHNIQUE

Definition: In two-way (full-duplex) communication, instead of sending an ACK packet all by itself, the receiver attaches (piggybacks) the ACK onto its next outgoing data frame.

Real-Life Example Delivering a package to your neighbor and returning their borrowed tool in the exact same visit instead of making two separate trips!

5. QUICK STUDENT REVIEW QUESTIONS

Q1: Why is Stop-and-Wait protocol inefficient on high-speed satellite links?

Ans: Because long propagation delays force the sender to sit idle waiting for the ACK, wasting high-speed channel capacity.

Q2: In Go-Back-N, if frames 1, 2, 3, and 4 are sent and frame 2 is lost, which frames are retransmitted?

Ans: Frames 2, 3, and 4 are all retransmitted, because the receiver discards all out-of-order frames after frame 2.